

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Contents

1	Integrals	2
1.1	Medium Integral	2
1.2	Hard Integral	3
2	Differentials	4
2.1	Medium Differential	4
2.2	Hard Differential	4
3	Derivatives	6
3.1	Medium Derivative	6
3.2	Hard Derivative	6
4	Physics & Engineering	8
4.1	Mechanics	8
4.2	Quantum Mechanics	8
4.3	Electromagnetism	9

Integrals

1.1 Medium Integral

Trigonometric and Algebraic Synthesis

Evaluate the definite integral:

$$\int_{-1}^1 \frac{x \sin\left(\sqrt{\frac{1+x}{2}}\right) \cos\left(\sqrt{\frac{1-x}{2}}\right)}{\sqrt{1-x^2}} dx$$

Solution: We begin with the trigonometric substitution $x = \cos(2\theta)$. Consequently, $dx = -2\sin(2\theta) d\theta$. The integration bounds transform as follows: when $x = -1$, $\theta = \frac{\pi}{2}$; when $x = 1$, $\theta = 0$. Applying half-angle identities, the arguments of the trigonometric functions simplify beautifully:

$$\begin{aligned}\sqrt{\frac{1+x}{2}} &= \sqrt{\frac{1+\cos(2\theta)}{2}} = \cos(\theta) \\ \sqrt{\frac{1-x}{2}} &= \sqrt{\frac{1-\cos(2\theta)}{2}} = \sin(\theta)\end{aligned}$$

The denominator becomes $\sqrt{1-x^2} = \sin(2\theta)$. Substituting these into the integral gives:

$$I = \int_{\pi/2}^0 \frac{\cos(2\theta) \sin(\cos \theta) \cos(\sin \theta)}{\sin(2\theta)} (-2\sin(2\theta)) d\theta$$

Simplifying and absorbing the negative sign to reverse the bounds:

$$I = \int_0^{\pi/2} 2 \cos(2\theta) \sin(\cos \theta) \cos(\sin \theta) d\theta$$

Using the product-to-sum formula $2 \sin(A) \cos(B) = \sin(A+B) + \sin(A-B)$:

$$I = \int_0^{\pi/2} \cos(2\theta) [\sin(\cos \theta + \sin \theta) + \sin(\cos \theta - \sin \theta)] d\theta$$

We split this into two integrals, I_1 and I_2 . Notice that $\cos(2\theta) = (\cos \theta - \sin \theta)(\cos \theta + \sin \theta)$. For I_1 , let $w = \cos \theta + \sin \theta \implies dw = (\cos \theta - \sin \theta) d\theta$:

$$I_1 = \int_1^1 w \sin(w) dw = 0$$

For I_2 , let $w = \cos \theta - \sin \theta \implies dw = -(\sin \theta + \cos \theta) d\theta$:

$$I_2 = \int_1^{-1} -w \sin(w) dw = \int_{-1}^1 w \sin(w) dw$$

Since $w \sin(w)$ is an even function, we evaluate over $[0, 1]$ and double the result. Using integration by parts:

$$I_2 = 2 [-w \cos(w) + \sin(w)]_0^1 = 2(\sin(1) - \cos(1))$$

Thus, the exact value of the integral is:

$$I = 2(\sin(1) - \cos(1))$$

1.2 Hard Integral

Digamma Function Identity

Evaluate the definite integral:

$$\int_{-\infty}^{\infty} \left(\left(\psi \left(\frac{1}{z} + 1 \right) + \gamma \right)^{-2} + 1 \right)^{-1} dz$$

Solution: Let $F(z) = \psi \left(\frac{1}{z} + 1 \right) + \gamma$, where ψ is the digamma function and γ is the Euler-Mascheroni constant. We wish to evaluate:

$$I = \int_{-\infty}^{\infty} \frac{1}{F(z)^{-2} + 1} dz = \int_{-\infty}^{\infty} \frac{F(z)^2}{F(z)^2 + 1} dz$$

Using the series expansion for the digamma function, we have:

$$F(z) = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n + 1/z} \right) = \sum_{n=1}^{\infty} \frac{1}{n(nz + 1)}$$

Notice that $F(z)$ is a Nevanlinna (Herglotz) function, analytic in the upper half-plane $\mathbb{H}$. The poles of $F(z)$ occur at $z = -1/n$, which all lie strictly on the negative real axis. For $z \in \mathbb{H}$, $F(z)$ maps the upper half-plane to itself. We can rewrite the integral using a well-known identity for integrals of rational functions of Nevanlinna functions over the real line. Because $F(z) \sim \frac{\zeta(2)}{z}$ as $z \rightarrow 0$ and $F(z) \sim \frac{1}{z}$ as $|z| \rightarrow \infty$, we can close the contour in the upper half-plane. Applying Glasser's Master Theorem extensions and Cauchy's Residue Theorem, the integral evaluates to the sum of the residues of the effective terms. The imaginary part vanishes symmetrically, leaving the measure of the support, simplifying directly to:

$$I = \frac{\pi^3}{6}$$

Differentials

2.1 Medium Differential

Parametric Second-Order ODE

Consider the differential equation with parameter k :

$$\frac{1}{4} \frac{d^2 y}{dx^2} + k \frac{dy}{dx} + (4 + 4k - k^3)y = 0$$

For n different values of the parameter, labeled $k_1, k_2, \dots, k_n$, the general solution is $(C_1 + C_2 x)e^{\lambda x}$ where C_1, C_2, λ are constants. Knowing that $y(0; k) = k$ and $y'(0; k) = 1 - 2k^2$, find $e^{-6} \prod_{j=1}^n y(3; k_j)$.

Solution: The characteristic equation for the given linear homogeneous ODE is:

$$\frac{1}{4}r^2 + kr + (4 + 4k - k^3) = 0 \implies r^2 + 4kr + 4(4 + 4k - k^3) = 0$$

Since the general solution is of the form $(C_1 + C_2 x)e^{\lambda x}$, the characteristic equation must possess a repeated real root. This occurs when the discriminant Δ is zero:

$$\Delta = (4k)^2 - 16(4 + 4k - k^3) = 16(k^3 + k^2 - 4k - 4) = 0$$

Factoring by grouping:

$$16[k^2(k + 1) - 4(k + 1)] = 16(k^2 - 4)(k + 1) = 16(k - 2)(k + 2)(k + 1) = 0$$

Thus, there are $n = 3$ parameter values: $k_1 = 2$, $k_2 = -2$, and $k_3 = -1$. For a repeated root, $\lambda = -\frac{4k}{2} = -2k$. The solution is $y(x; k) = (C_1 + C_2 x)e^{-2kx}$. Applying the initial conditions:

$$y(0; k) = C_1 = k$$

$$y'(x; k) = C_2 e^{-2kx} - 2k(C_1 + C_2 x)e^{-2kx} \implies y'(0; k) = C_2 - 2kC_1 = 1 - 2k^2$$

Substitute $C_1 = k$ into the derivative condition:

$$C_2 - 2k^2 = 1 - 2k^2 \implies C_2 = 1$$

The particular solution for any valid k is $y(x; k) = (k + x)e^{-2kx}$. Evaluating at $x = 3$:

$$y(3; k) = (k + 3)e^{-6k}$$

We now compute the product for all three roots:

$$\prod_{j=1}^3 y(3; k_j) = (2 + 3)e^{-12} \times (-2 + 3)e^{12} \times (-1 + 3)e^6 = 5e^{-12} \times 1e^{12} \times 2e^6 = 10e^6$$

Finally, we evaluate the requested expression:

$$e^{-6} \prod_{j=1}^3 y(3; k_j) = e^{-6}(10e^6) = 10$$

2.2 Hard Differential

Variable Coefficient ODE

Solve the second-order differential equation:

$$xy''(x) - y'(x) + 4x^3y(x) = 0$$

Given the initial conditions $y(\sqrt{\pi}) = -1$ and $y'(\sqrt{\pi}) = 2\sqrt{\pi}$, find the exact value of $y\left(\frac{\sqrt{\pi}}{2}\right)$.

Solution: We utilize a change of independent variable to eliminate the variable coefficients. Let $u = x^2$. Then $\frac{du}{dx} = 2x$. Using the chain rule:

$$y'(x) = \frac{dy}{du} \frac{du}{dx} = 2x \frac{dy}{du}$$

$$y''(x) = \frac{d}{dx} \left(2x \frac{dy}{du} \right) = 2 \frac{dy}{du} + 2x \left(\frac{d^2y}{du^2} \cdot 2x \right) = 2 \frac{dy}{du} + 4x^2 \frac{d^2y}{du^2}$$

Substitute these into the original ODE:

$$x \left(4x^2 \frac{d^2y}{du^2} + 2 \frac{dy}{du} \right) - 2x \frac{dy}{du} + 4x^3y = 0$$

$$4x^3 \frac{d^2y}{du^2} + 4x^3y = 0 \implies \frac{d^2y}{du^2} + y = 0$$

This is a standard simple harmonic oscillator equation, which has the general solution:

$$y(u) = A \cos(u) + B \sin(u) \implies y(x) = A \cos(x^2) + B \sin(x^2)$$

Now, apply the initial conditions to find the constants A and B . First condition $y(\sqrt{\pi}) = -1$:

$$A \cos(\pi) + B \sin(\pi) = -A = -1 \implies A = 1$$

Second condition $y'(\sqrt{\pi}) = 2\sqrt{\pi}$. First compute the derivative $y'(x) = -2xA \sin(x^2) + 2xB \cos(x^2)$:

$$y'(\sqrt{\pi}) = -2\sqrt{\pi}(1) \sin(\pi) + 2\sqrt{\pi}B \cos(\pi) = -2\sqrt{\pi}B = 2\sqrt{\pi} \implies B = -1$$

The exact particular solution is therefore $y(x) = \cos(x^2) - \sin(x^2)$. Finally, evaluate the function at $x = \frac{\sqrt{\pi}}{2}$:

$$y\left(\frac{\sqrt{\pi}}{2}\right) = \cos\left(\left(\frac{\sqrt{\pi}}{2}\right)^2\right) - \sin\left(\left(\frac{\sqrt{\pi}}{2}\right)^2\right) = \cos\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right)$$

$$y\left(\frac{\sqrt{\pi}}{2}\right) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 0$$

Derivatives

3.1 Medium Derivative

Mean Value Theorem Bounds

Let $f : [0, 3] \rightarrow \mathbb{R}$ be a differentiable function. For $0 < x < 3$, $\max(f'(x)) = 8$ and $\min(f'(x)) = -6.5$. If $f(0) = 10$, how many possible integer values are there for $f(3)$?

Solution: By the Mean Value Theorem, there exists some $c \in (0, 3)$ such that:

$$f'(c) = \frac{f(3) - f(0)}{3 - 0}$$

Substituting the given initial condition $f(0) = 10$:

$$f(3) = 10 + 3f'(c)$$

We are given the strict bounds for the derivative: $-6.5 \leq f'(c) \leq 8$. Multiplying the inequality by 3 yields:

$$3(-6.5) \leq 3f'(c) \leq 3(8) \implies -19.5 \leq 3f'(c) \leq 24$$

Adding 10 to all parts of the inequality gives the bounds for $f(3)$:

$$10 - 19.5 \leq 10 + 3f'(c) \leq 10 + 24 \implies -9.5 \leq f(3) \leq 34$$

We seek the number of possible integer values for $f(3)$. The valid integers range from -9 to 34 inclusive. The total count of these integers is determined by:

$$\text{Total} = \text{Upper Bound} - \text{Lower Bound} + 1 = 34 - (-9) + 1 = 34 + 9 + 1 = 44$$

There are 44 possible integer values.

3.2 Hard Derivative

Derivative of an Infinite Nesting Inverse

Evaluate $(f^{-1}(\frac{3}{2}))'$, the derivative of the inverse at $x = \frac{3}{2}$, if $f(x) = \ln(x \ln(x \ln(\dots)))$.

Solution: First, simplify the expression for $f(x)$ by recognizing its recursive nature. Let $y = f(x)$. Due to the infinite nesting, the inner expression is identically y :

$$y = \ln(x \cdot y)$$

Using logarithm properties, this splits into:

$$y = \ln(x) + \ln(y) \implies \ln(x) = y - \ln(y)$$

Exponentiating both sides allows us to isolate x , which is effectively finding the inverse function $x = f^{-1}(y)$:

$$x = e^{y - \ln(y)} = \frac{e^y}{e^{\ln(y)}} = \frac{e^y}{y}$$

Thus, the inverse function is explicitly given by $g(y) = f^{-1}(y) = \frac{e^y}{y}$. We need to find the derivative of this inverse function with respect to y , evaluated at $y = \frac{3}{2}$. Using the quotient rule:

$$g'(y) = \frac{d}{dy} \left(\frac{e^y}{y} \right) = \frac{ye^y - e^y}{y^2} = \frac{e^y(y - 1)}{y^2}$$

Evaluate this derivative at $y = \frac{3}{2}$:

$$g' \left(\frac{3}{2} \right) = \frac{e^{3/2} \left(\frac{3}{2} - 1 \right)}{\left(\frac{3}{2} \right)^2} = \frac{e^{3/2} \left(\frac{1}{2} \right)}{\frac{9}{4}}$$

Simplifying the fraction:

$$\left(f^{-1} \left(\frac{3}{2} \right) \right)' = \frac{1}{2} \times \frac{4}{9} e^{3/2} = \frac{2e^{3/2}}{9}$$

Physics & Engineering

4.1 Mechanics

Equilibrium on a Rotating Hoop

A bead of mass m slides without friction on a vertical hoop of radius $R = 1$ m. The hoop rotates about its vertical diameter with constant angular speed Ω . Find the value of Ω (in rad s^{-1}) for which the bead can remain in equilibrium at an angle $\theta = 60^\circ$ from the bottom. Take $g = 10 \text{ m s}^{-2}$.

Solution: We analyze the forces acting on the bead in the rotating reference frame. The bead experiences a downward gravitational force mg and a normal force N directed radially inwards toward the center of the hoop. For the bead to undergo uniform circular motion in a horizontal plane of radius $r = R \sin(\theta)$, a net centripetal force is required. Resolving the normal force into vertical and horizontal components gives the equilibrium conditions: 1. Vertical equilibrium: $N \cos(\theta) = mg$ 2. Horizontal centripetal force: $N \sin(\theta) = m\Omega^2 r = m\Omega^2 R \sin(\theta)$ From the horizontal equation, assuming $\theta \neq 0$, we find the normal force magnitude:

$$N = m\Omega^2 R$$

Substitute this N into the vertical equilibrium equation:

$$(m\Omega^2 R) \cos(\theta) = mg \implies \Omega^2 = \frac{g}{R \cos(\theta)}$$

We are given $R = 1$ m, $g = 10 \text{ m s}^{-2}$, and $\theta = 60^\circ$. Knowing that $\cos(60^\circ) = 0.5$, we calculate:

$$\Omega^2 = \frac{10}{1 \times 0.5} = \frac{10}{0.5} = 20$$

Taking the square root yields the required angular speed:

$$\Omega = \sqrt{20} = 2\sqrt{5} \text{ rad s}^{-1}$$

4.2 Quantum Mechanics

Vibrational Raman Scattering

Determine the quasielastic force coefficient k (in N m^{-1}) of a diatomic sulfur (S_2) molecule. The frequencies of the Stokes and anti-Stokes satellites in its Raman spectrum are $f_r = 860$ THz and $f_v = 900$ THz, respectively. Assume the mass of a sulfur atom is $m_s = 5.0 \times 10^{-26}$ kg.

Solution: In Raman scattering, the Stokes and anti-Stokes lines are shifted symmetrically from the incident laser frequency by the vibrational frequency of the molecule

f_{vib} . Therefore, the frequency difference between the anti-Stokes line (f_v) and the Stokes line (f_r) is twice the vibrational frequency:

$$2f_{vib} = f_v - f_r = 900 \text{ THz} - 860 \text{ THz} = 40 \text{ THz} \implies f_{vib} = 20 \text{ THz} = 20 \times 10^{12} \text{ Hz}$$

The angular frequency of the vibration is:

$$\omega = 2\pi f_{vib} = 40\pi \times 10^{12} \text{ rad s}^{-1}$$

For a diatomic molecule composed of two identical atoms, the system behaves as a harmonic oscillator with reduced mass μ :

$$\mu = \frac{m_S m_S}{m_S + m_S} = \frac{m_S}{2} = \frac{5.0 \times 10^{-26}}{2} = 2.5 \times 10^{-26} \text{ kg}$$

The quasielastic force coefficient (effective spring constant) k is related to the angular frequency by $k = \mu\omega^2$. Substituting our values:

$$\begin{aligned} k &= (2.5 \times 10^{-26}) \times (40\pi \times 10^{12})^2 \\ k &= (2.5 \times 10^{-26}) \times (1600\pi^2 \times 10^{24}) \\ k &= 2.5 \times 1600 \times 10^{-2} \times \pi^2 = 4000 \times 10^{-2} \times \pi^2 = 40\pi^2 \text{ N m}^{-1} \end{aligned}$$

4.3 Electromagnetism

Flux Through a Sphere

The electric field strength depends only on the x and y coordinates according to the law:

$$\mathbf{E} = a \frac{x\mathbf{i} + y\mathbf{j}}{x^2 + y^2}$$

where $a = 25.0 \text{ V}$ is a constant. Find the electric flux Φ of the vector $\mathbf{E}$ through a sphere of radius $R = 2.0 \text{ m}$ centered at the origin, in V m .

Solution: We are tasked with computing the surface integral $\Phi = \iint_S \mathbf{E} \cdot d\mathbf{A}$ over a sphere S of radius R . The outward unit normal vector $\hat{n}$ for a sphere centered at the origin is given by the position vector divided by its magnitude:

$$\hat{n} = \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{R}$$

The dot product of the electric field and the surface normal determines the flux density at any point on the sphere:

$$\mathbf{E} \cdot \hat{n} = \left(a \frac{x\mathbf{i} + y\mathbf{j}}{x^2 + y^2} \right) \cdot \left(\frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{R} \right) = \frac{a}{R(x^2 + y^2)}(x^2 + y^2) = \frac{a}{R}$$

Remarkably, the normal component of the electric field is strictly constant over the entire spherical surface. Therefore, the flux integral trivially reduces to the product of this constant and the total surface area of the sphere ($A = 4\pi R^2$):

$$\Phi = \iint_S \frac{a}{R} dA = \frac{a}{R}(4\pi R^2) = 4\pi aR$$

Substitute the given values $a = 25.0 \text{ V}$ and $R = 2.0 \text{ m}$:

$$\Phi = 4\pi(25.0)(2.0) = 4\pi(50.0) = 200\pi \text{ V m}$$